

JKBOSE

2026 CLASS 11TH

MOTION IN PLANE

LECTURE-105

PHYSICS

WAJAHID SIR (Gabbar Sir)



Topics To Covered

1 Revision ✓

2 *How to solve problems*

3 ..

4 ..

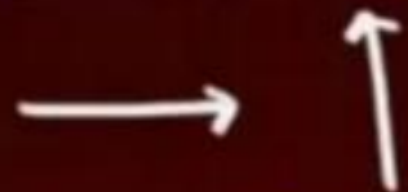
One Short

10 min

Motion in 2D (Revision)

Recap from 1st Lecture

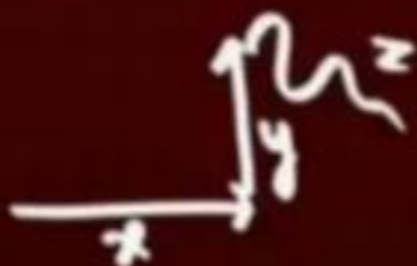
- Motion in 1D



- Motion in 2D



- Motion in 3D



- Displacement in 2D

$$\vec{r} = \vec{x} + \vec{y}$$

- Velocity in 2D

$$\vec{v} = \vec{v}_x + \vec{v}_y$$

- Acceleration in 2D

$$\vec{a} = \vec{a}_x + \vec{a}_y$$

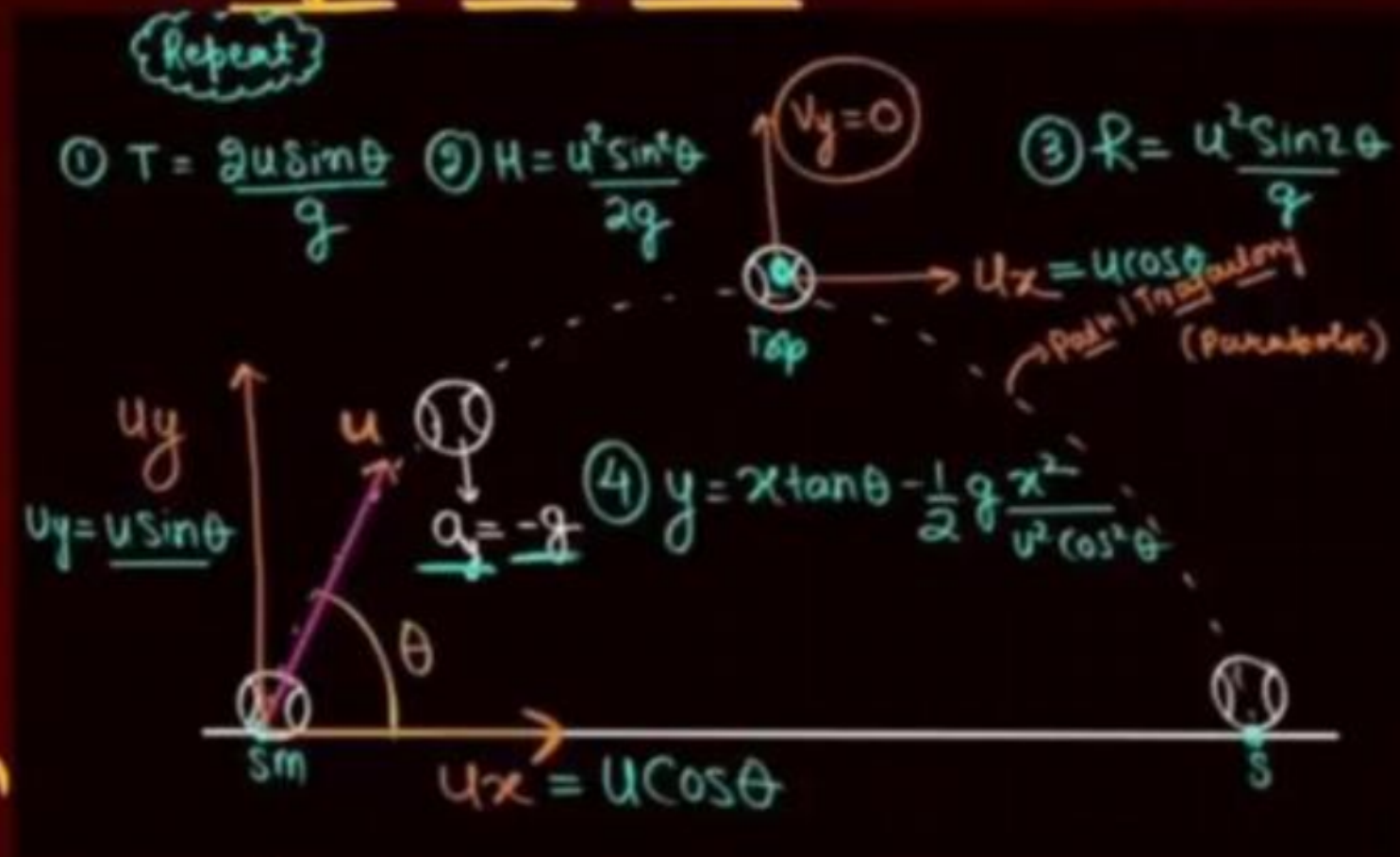
- Equation of motion

$$v = u + at \quad s = ut + \frac{1}{2}at^2 \quad v^2 = u^2 + 2as$$

$$v_x = u_x + a_x t \quad s_x = u_x t + \frac{1}{2}a_x t^2 \quad v_x^2 = u_x^2 + 2a_x s_x$$

$$v_y = u_y + a_y t \quad s_y = u_y t + \frac{1}{2}a_y t^2 \quad v_y^2 = u_y^2 + 2a_y s_y$$

Projectile Motion



Projectile Motion

(5-mark) 2026, 2025, 2024, 2023, 2022



- Example of 2D
- To explain the 2D motion
- when an object is thrown with Some Angle under the influence of gravity

Neet level

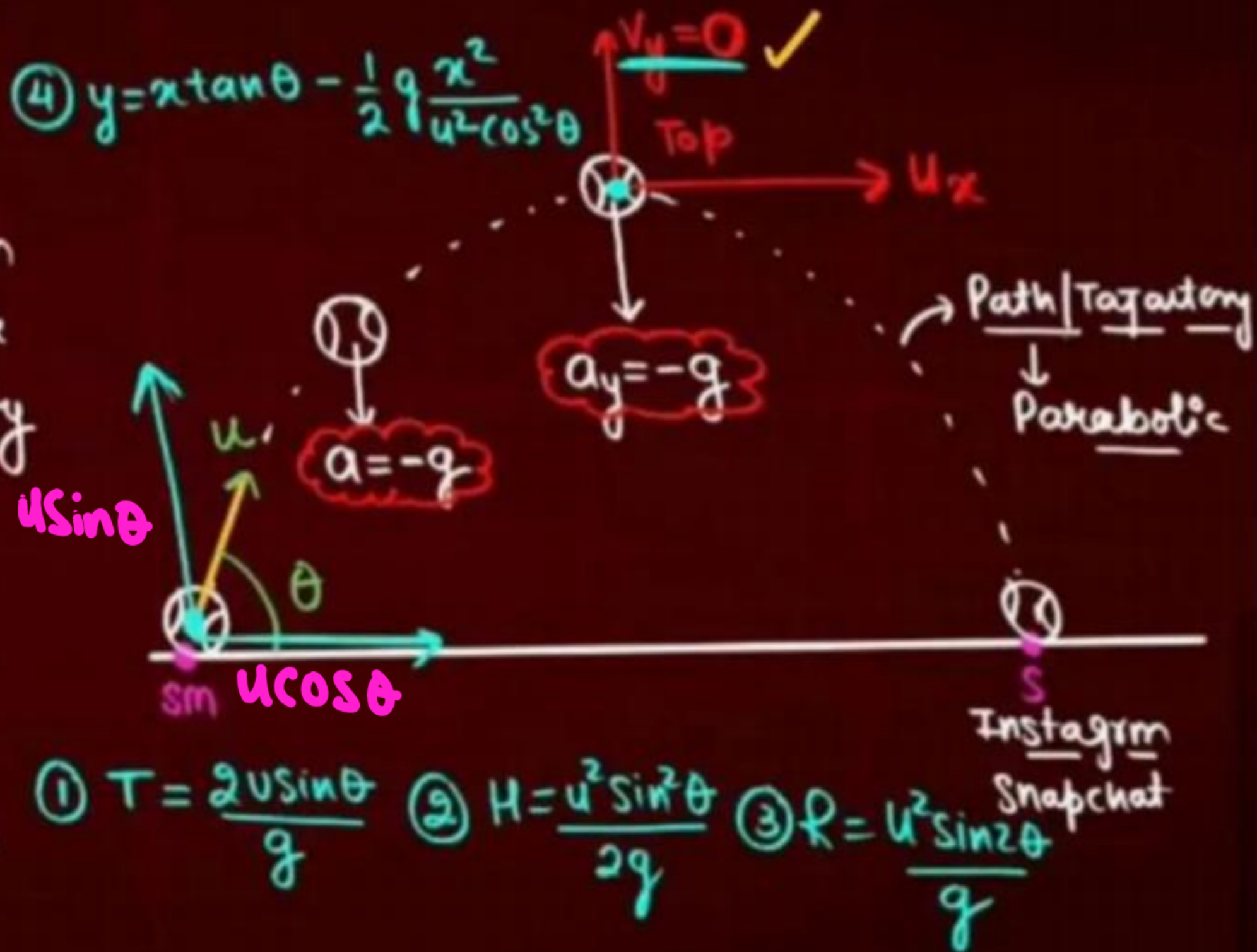
$a_y = -g = -9.8 = -10 \text{ m/s}^2$

$a_x = 0$

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$v_y = 0$

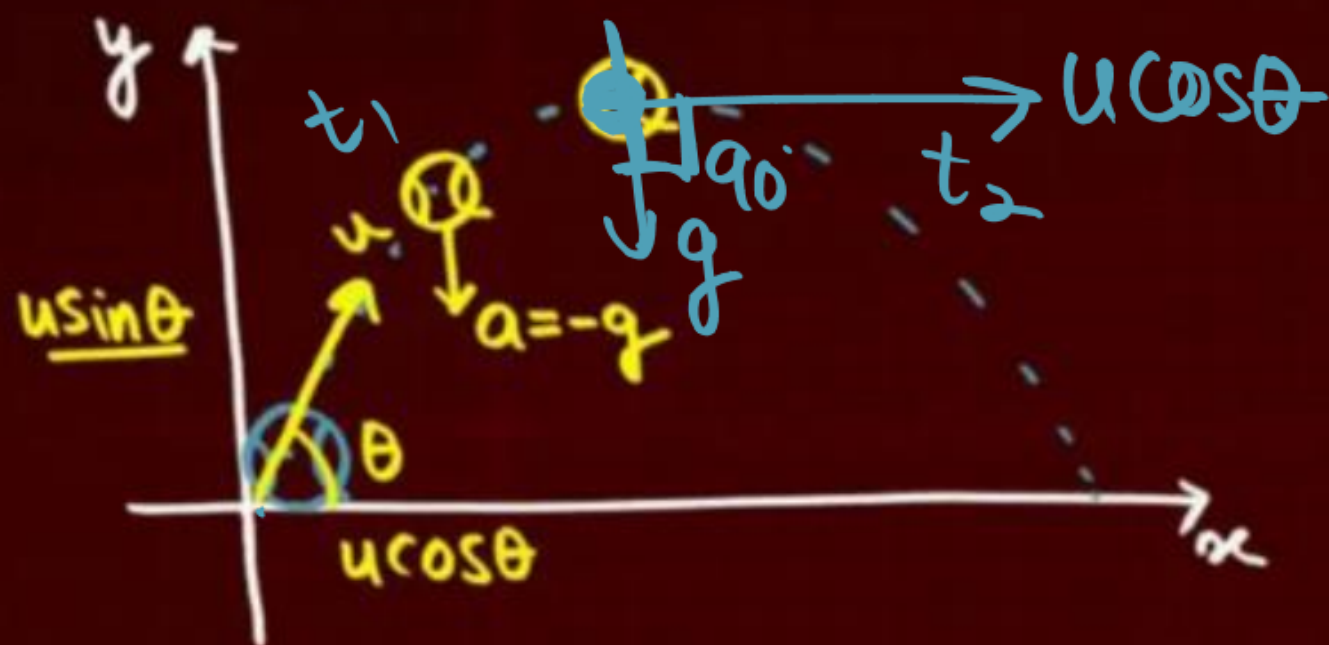
Path | Trajectory $\rightarrow$ Parabolic



Recap from 2nd Lecture



Projectile Motion



① Time period. $t_1 = t_2$

The time for which ball remain in Air $T = \frac{2u \sin \theta}{g}$

② Maximum Height.

The maximum height attain by Projectile

$$H = \frac{u^2 \sin^2 \theta}{2g}$$

③ Range

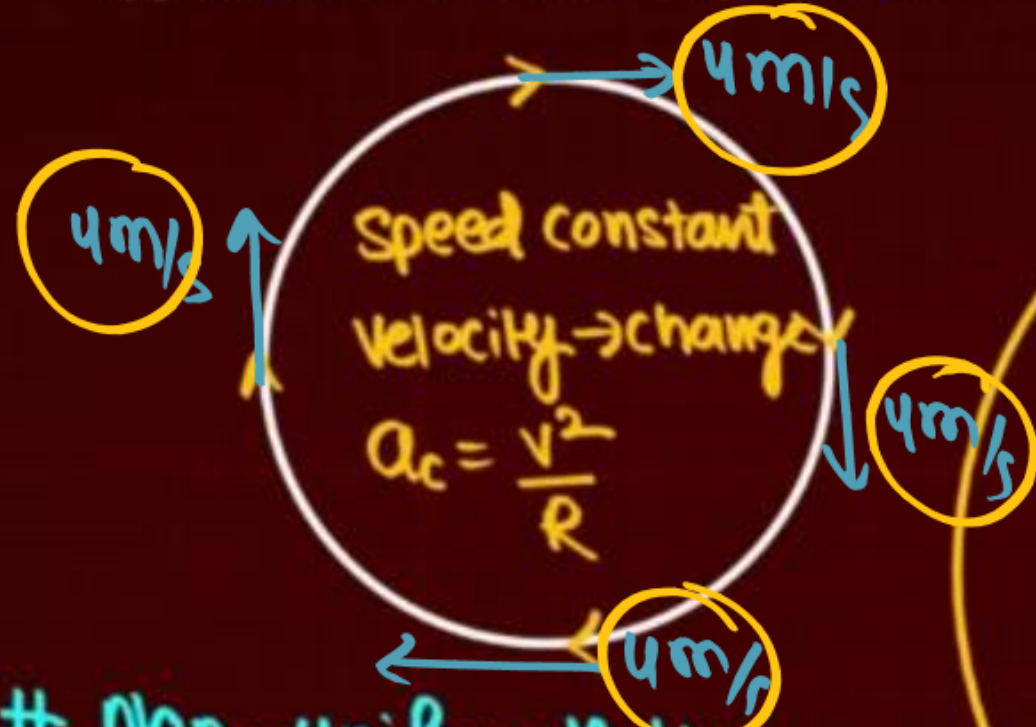
The maximum Horizontal displacement of object $R = \frac{u^2 \sin 2\theta}{g}$

④ Equation of Trajectory $\rightarrow$ Relationship b/w y and x

$$y = x \tan \theta - \frac{1}{2} g \frac{x^2}{u^2 \cos^2 \theta}$$

Recap from 3rd Lecture

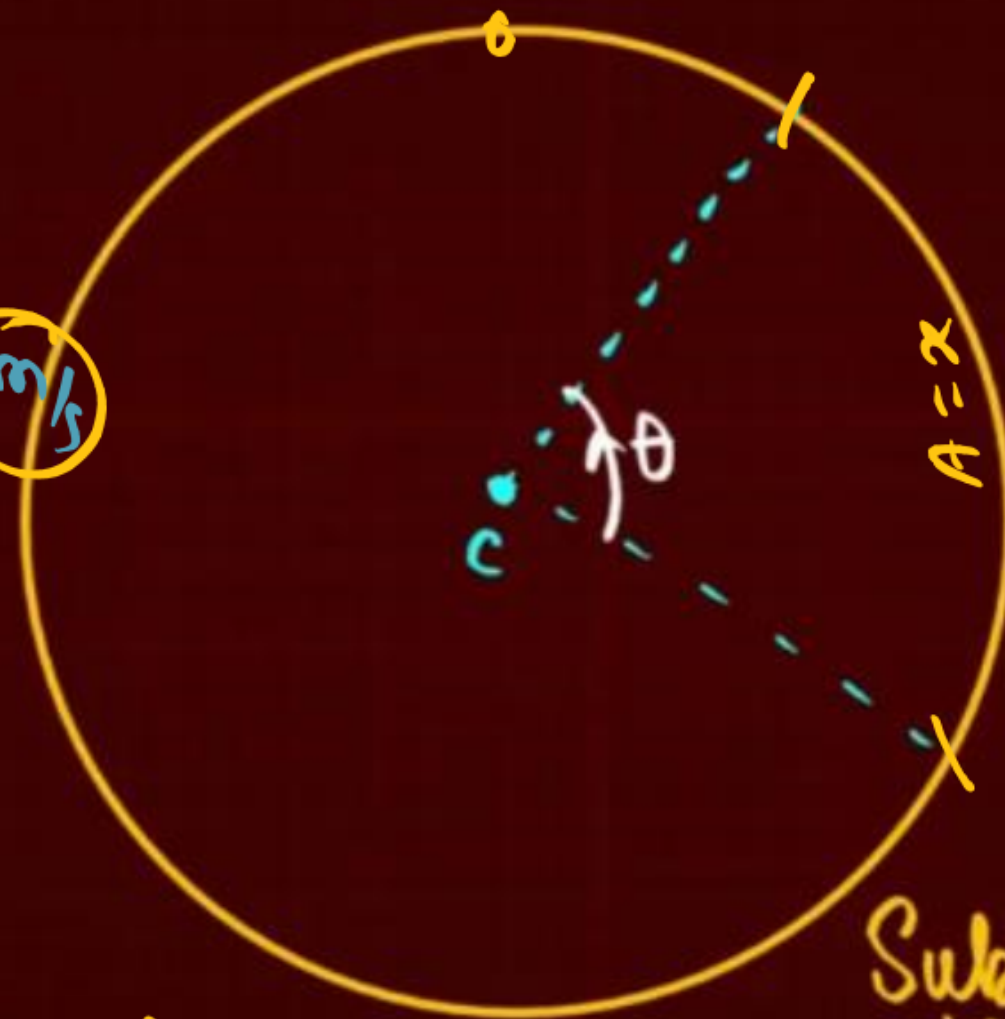
Uniform Circular Motion



Non-uniform Motion

Speed = $\frac{2\pi r}{T} = 2\pi r \gamma$

$$\begin{matrix} A+B \\ \vec{A} - \vec{B} \end{matrix}$$



① Angular displacement

$$\theta = \frac{A}{r}$$

$$\gamma \theta = \alpha$$

$$\theta = \frac{\alpha}{\gamma}$$

② Angular velocity (ω)

$$\gamma \omega = v$$

③ Angular acceleration (α)

$$\gamma \alpha = a$$

Subtraction of two vector

$$R = \sqrt{A^2 + B^2 - 2AB \cos \theta} \quad \tan \alpha = \frac{B \sin \theta}{A - B \cos \theta}$$

R R

chapter complete

Thank
You

✓
Revision → 10 min

Notes → There's a
late